3. Functional Requirements

Each requirement is assigned a unique ID for traceability, grouped by functional module. Priority is expressed as High, Medium, or Low.

3.1 User Registration & Authentication

ID	Requirement	Priority
FR-AUTH-1	The system shall allow a new student to register by providing full name, student ID number, department, semester, and phone number.	High
FR-AUTH-2	The system shall automatically generate a unique login name from the student's full name, appending a numeric suffix (e.g., "2", "3") when the name is already registered.	High
FR-AUTH-3	The system shall assign a default password of "1234" to every newly registered student account.	High
FR-AUTH-4	The system shall allow a student to log in using their generated login name and password.	High
FR-AUTH-5	The system shall display live name suggestions, filtered from the first character typed, while a student enters their login name.	High
FR-AUTH-6	The system shall allow the administrator to log in using only a password, without a separate identifier field.	High
FR-AUTH-7	The system shall issue a signed JWT session token upon successful login, valid for 12 hours.	High
FR-AUTH-8	The system shall allow a logged-in user (student or admin) to change their password at any time from within their portal, without forcing this action immediately after login.	Medium
FR-AUTH-9	The system shall reject login attempts with incorrect credentials and return a generic error without revealing whether the identifier or password was invalid.	Medium

3.2 Landing Page & Student Directory

ID	Requirement	Priority
FR-LAND-1	The system shall present a single-block landing page featuring the project logo and three primary options: Login as Student, Login as Admin, and All Posts.	High

ID	Requirement	Priority
FR-LAND-2	The system shall provide an information box on the landing page that, when activated, displays a searchable directory of all registered students.	High
FR-LAND-3	The directory shall display each student's name, student ID, department, and semester, excluding sensitive contact details.	High
FR-LAND-4	The system shall filter the directory results live as the user types, matching on name, student ID, or department, with name-based suggestions.	High

3.3 Lost Item Reporting

ID	Requirement	Priority
FR-LOST-1	The system shall allow a logged-in student to report a lost item by specifying the date lost, location, category, size, brand/model, color, and a free-text description.	High
FR-LOST-2	The system shall present location and category fields as dropdown-with-suggestions inputs populated from a predefined BUP-specific reference list.	High
FR-LOST-3	The system shall dynamically populate size and color options based on the category selected.	Medium
FR-LOST-4	The system shall publicly display the reporting student's name, phone number, and department alongside a lost item post.	High
FR-LOST-5	The system shall allow a student to view and delete their own active lost item posts.	Medium
FR-LOST-6	The system shall prevent deletion of a lost item post once it has been marked as returned.	Medium

3.4 Found Item Reporting

ID	Requirement	Priority
FR-FOUND-1	The system shall allow a logged-in student to report a found item by specifying the date found, location, category, size, and brand/model.	High
FR-FOUND-2	The system shall collect additional admin-only fields (color and a free-text description) that are never shown on the public post.	High

ID	Requirement	Priority
FR-FOUND-3	The system shall generate a unique, sequential token number for every found item post.	High
FR-FOUND-4	The system shall keep the identity of the finder hidden from the public post; only the administrator and the finder may view it.	High
FR-FOUND-5	The system shall allow a student to view their own found item posts along with the associated token numbers.	Medium

3.5 Browsing & Search

ID	Requirement	Priority
FR-BROWSE-1	The system shall allow any visitor, logged in or not, to browse all active lost item posts.	High
FR-BROWSE-2	The system shall allow any visitor, logged in or not, to browse all active found item posts (with finder identity withheld).	High
FR-BROWSE-3	The system shall allow browsing results to be filtered by item category.	Medium
FR-BROWSE-4	The system shall allow any visitor to view the archive of successfully returned items.	Low

3.6 Administrator Portal & Category Management

ID	Requirement	Priority
FR-ADMIN-1	The system shall display, for each item category, the current count of active lost posts and active found posts.	High
FR-ADMIN-2	The system shall allow the administrator to open a category and view all found items (left) and all lost items (right) within it.	High
FR-ADMIN-3	The system shall display finder and owner contact details to the administrator within the category matching workspace.	High
FR-ADMIN-4	The system shall allow the administrator to confirm a match between a specific found item and a specific lost item.	High
FR-ADMIN-5	Upon confirmation, the system shall mark both the lost and found posts as "returned" and move them out of the active listings.	High

3.7 Matching Algorithm

ID	Requirement	Priority
FR-MATCH-1	The system shall compute a match score between a found item and a lost item only when both belong to the same category.	High
FR-MATCH-2	The system shall weight the match score using color similarity (25 points), size match (15 points), brand/model similarity (15 points), same location (15 points), date proximity (10 points), and free-text description keyword overlap (20 points).	High
FR-MATCH-3	The system shall present candidate lost items for a given found item ranked from highest match score to lowest.	High
FR-MATCH-4	The system shall require explicit administrator confirmation before finalizing any match; the score is advisory, not automatic.	High

3.8 Returned Items & Notifications

ID	Requirement	Priority
FR-RETURN-1	The system shall record every confirmed return in a permanent, publicly viewable archive including the matched category, score, owner, and finder.	High
FR-RETURN-2	The system shall generate an in-app notification for the item’s owner immediately upon a confirmed return, including the finder’s name, student ID, and phone number.	High
FR-RETURN-3	The system shall display a notification bell with an unread-count badge to logged-in users.	Medium
FR-RETURN-4	The system shall allow a user to mark an individual notification as read.	Low